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101
ALL

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Mathematics **Subject Code : BSC 101**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory.*

The figures in the margin indicate full marks.

Marks are given in the right margin.

1. Choose the correct answer in each of the following: 2×7=14

(i) If $2\begin{pmatrix}3 & 4\\ 5 & x\end{pmatrix}+\begin{pmatrix}1 & 2y\\ 0 & 1\end{pmatrix}=\begin{pmatrix}7 & 0\\ 10 & 5\end{pmatrix}$, then values of x and y are

(a) $x=8, y=0$ (b) $x=2, y=-4$

(c) $x=6, y=2$ (d) $x=5, y=3$

(ii) If $\begin{pmatrix}2x+y\\ x-y\end{pmatrix}=\begin{pmatrix}8\\ 4\end{pmatrix}$, then values of x and y are

(a) $x=4, y=0$ (b) $x=8, y=0$

(c) $x=6, y=2$ (d) $x=5, y=3$

(iii) The distance between the points $A(2,-7)$ and $B(6,3)$ is

(a) 4 units (b) 6 units

(c) 8 units (d) 10 units

(iv) The distance between the points $A(2,-3)$ and $B(-6,3)$ is

(a) $\sqrt{29}$ units (b) $-\sqrt{29}$ units

(c) $4\sqrt{29}$ units (d) $2\sqrt{29}$ units

(v) If the slope of the line joining the points $A(x,2)$ and $B(6,-8)$ is $\frac{-3}{4}$, then $x=?$

(a) $\frac{-22}{3}$ (b) $\frac{22}{3}$

(c) $\frac{-2}{3}$ (d) 3

(vi) $\sin\left(\frac{19\pi}{3}\right)=?$

(a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

(vii) $\operatorname{cosec} 840° = ?$

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

2. (a) Solve the following system of linear equations (use any method):

$$3x + 4y + 2z = 8$$
$$x - 2y - 3z = 3$$
$$3x - 2y + 6z = -2$$

(b) Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. 7+7

3. (a) Find the equation of the line whose portion intercepted between the axes is bisected at the point $(3, -2)$.

(b) Find the equation of the line whose y-intercept is -3 and which is perpendicular to the line joining the points $(-2, 3)$ and $(4, -5)$. 7+7

4. (a) If $\sin\theta = \frac{\sqrt{3}}{2}$ and θ lies in quadrant II, find the values of all the other five trigonometric functions.

(b) Calculate the values of: (i) $\sin 15°$, (ii) $\cos 15°$, (iii) $\tan 15°$. 7+7

5. (a) Differentiate y $w.r.t.$ x:

$$y = \log\left(\operatorname{cosec}\frac{x}{2} + \cot\frac{x}{2}\right)$$

(b) Find the point of local maxima or minima and corresponding local maximum and minimum value for the following function:

$$f(x) = -x^3 + 12x^2 - 5$$ 7+7

6. (a) Evaluate the integral: $I = \int_0^{\frac{\pi}{2}} \cos^4 x dx$.

(b) Evaluate the integral: $I = \int \left(\frac{x^3+4x^2-3x-2}{x+2}\right) dx$ 7+7

7. Answer *any four* of the following: 3.5×4=14

(a) If $2\begin{pmatrix} x & 5 \\ 7 & y-3 \end{pmatrix} + \begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ 15 & 14 \end{pmatrix}$, then find the value of x and y.

(b) Find the equation of the line perpendicular to the line $x - 7y + 5 = 0$ and having x-intercept 3.

(c) Prove that $\left(\frac{\cos x+\sin x}{\cos x-\sin x} - \frac{\cos x-\sin x}{\cos x+\sin x}\right) = 2\tan 2x$.

(d) A spherical soap bubble is expanding so that its radius is increasing at the rate of 0.02 cm/sec. At what rate is the surface area increasing when its radius is 5 cm? (Take $\pi = 3.14$)

(e) Evaluate the integral: $I = \int \left(\frac{2}{1+x^2} - \frac{2}{x^{\frac{3}{4}}} + 3x^{-1} - \frac{3}{x\sqrt{x^2-1}} + 4e^{\frac{x}{4}} + 5^x\right) dx$